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15 June 1981
SUPERSEDING
MIL-S-22999A(WEP)
3 February 1964

MILITARY SPECIFICATION

STARTERS, AIRCRAFT ENGINE, HYDRAULIC GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for aircraft turbine engine hydraulic starters.

1.2 Classification. Starters covered by this specification shall be Types I and II.

1.2.1 Types. Type I starters shall be designed to operate as a part of a dual pressure, nominal 3000-psi and 4000-psi hydraulic system. They shall incorporate a flow-limiting device and shall be capable of operating satisfactorily from power sources having unlimited flow and from power sources limited to flow as specified in the applicable specification sheet and herein. Type II starters shall be designed to operate as a part of a nominal 4000-psi flow-limited hydraulic system. Type I and II hydraulic starters shall be designed to conform to Type II hydraulic system components as defined in MIL-H-8775, and shall be designed to operate in hydraulic systems conforming to the requirements of MIL-H-5440.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ENGINEERING SPECIFICATIONS AND STANDARDS DEPARTMENT (CODE 93) NAVAL AIR ENGINEERING CENTER, LAKEHURST NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 2995

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SPECIFICATIONS

Federal

PPP-B-585	Box, Wood, Wirewound (for Domestic Shipment)
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-T-60	Tape, Packaging, Waterproof

Military

MIL-P-116	Preservation Packaging, Methods of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-H-5440	Hydraulic Systems, Aircraft, Types I and II, Design and Installation Requirements for
MIL-C-5501	Cap and Plug, Protective, Dust and Moisture Seal
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile and Ordnance
MIL-C-6021	Castings, Classification and Inspection of
MIL-H-6083	Hydraulic Fluid, Petroleum Base, for Preservation and Operation
MIL-H-6088	Heat Treatment of Aluminum Alloys
MIL-H-6875	Heat Treatment of Steels (Aircraft Practice), Process for
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements for
MIL-F-7179	Finishes and Coatings, Protection of Aerospace Weapons Systems, Structures and Parts, General Specification for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for

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SPECIFICATIONS (Continued)

Military (Continued)

MIL-S-7780	Starter, Electric, Constant Current, DC
MIL-S-7848	Starter, Engine, Air Turbine Type, General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-H-8775	Hydraulic System Components, Aircraft and Missiles, General Specification for
MIL-F-8815	Filter and Filter Elements, Fluid Pressure, Hydraulic Line, 15-Micron Absolute, and 5-Micron Absolute, Type II Systems, General Specification for
MIL-S-8879	Screw Threads, Controlled Radius with Increased Minor Diameter, General Specification for
MIL-C-11796	Corrosion Preventive Compound, Petrolatum, Hot Application
MIL-C-16173	Corrosion Preventive Compound, Solvent Cutback, Cold Application
MIL-A-21180	Aluminum-Alloy Castings, High Strength
MIL-S-22999/1	Starter, Aircraft Engine, Hydraulic, for T58 Engine
MIL-S-22999/2	Starter, Aircraft Engine, Hydraulic, for T64 Engine
MIL-H-25475	Hydraulic System, Missile, Design, Installation Tests, and Data Requirements, General Requirements for
MIL-H-83282	Hydraulic Fluid, Fire Resistant Synthetic Hydrocarbon Base, Aircraft

STANDARDS

Military

DOD-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property

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STANDARDS (Continued)

Military (Continued)

MIL-STD-280	Definitions of Item Levels, Item Exchangeability, Models and Related Terms
MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals
MIL-STD-1523	Age Control of Age Sensitive Elastomeric Material
MS14118	Adapter, Hydraulic Starter to Engine, Mounting
MS33540	Safety Wiring and Cotter Pinning, General Practices for
MS33649	Boss, Fluid Connection - Internal Straight Thread

Air Force - Navy Aeronautical

AND10050	Bosses, Standard Dimensions for Gasket Seal Straight Thread (Inactive for Design, Use MS33649)
AND20002	Drive, Type XII, Engine Accessory
AND20009	Drive, Type XIX, Engine Accessory

PUBLICATIONS

Air Force - Navy Aeronautical Bulletins

MIL-BULL-544	List of Specifications and Standards (Book Form) Approved by the Naval Air Systems Command

(When requesting applicable documents, refer to both title and number. Copies of unclassified documents may be obtained from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. Requests for copies of classified documents should be addressed to the Naval Publications and Forms Center, via the cognizant Government representative.)

American National Standards Institute (ANSI)

B46.1-1962	Surface Texture

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

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3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.1.1 Precedence. The contract shall have precedence over any specification. Any design or construction change that may effect compliance with this specification or from subsidiary specifications or standards, for a qualified starter, are subject to approval in writing by the qualifying activity (see 3.15.3).

3.2 Qualification. The starter furnished under this specification shall be a product which has been qualified for listing on the applicable Qualified Products List at the time set for opening of bids (see 4.2 and 6.3).

3.3 Dimensions. Each part shall conform to the dimensions shown on the specification sheet. Where only maximum or minimum dimensions are shown, the part need not have the shape shown, but the part, including all protrusions, shall be contained within the outline shown. Design details which are shown but are not specifically required are shown as examples and for information only; such details are optional.

3.4 Data. Unless otherwise specified in the contract, data cited in this specification is only required as part of the qualification submittal.

3.5 Materials and parts. Materials and parts used in construction of the starter shall be unaffected by hydraulic fluids (MIL-H-5606 or MIL-H-83282) while the starter is operating under any of the specified conditions. Materials, parts, and processes shall conform to applicable Government specifications listed in MIL-BULL-544. Contractor or industry specifications may be used for materials or processes not covered by the documents listed in MIL-BULL-544, provided the specifications are approved by the qualifying activity and contain provisions for adequate tests.

3.5.1 Materials.

3.5.1.1 Toxic products. Materials which are toxic in themselves, or which become toxic when pulverized or vaporized, or which have toxic products of combustion, shall be used only upon written permission of the qualifying activity.

3.5.1.2 Non-metals. Non-metallic materials used, including plastics, fabrics, and protective finishes shall be flame-resistant, shall not support combustion, shall be moisture resistant, and shall not be adversely affected by weather, temperature, and ambient conditions.

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3.5.1.2.1 Fungus resistance. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used, all nutrient materials and associated fungicidal treatments used shall be specified on the starter manufacturing drawings.

3.5.1.2.2 Packings. All packings shall be in accordance with the requirements of MIL-H-5440 or MIL-H-25475 as applicable. Age of synthetic rubber parts shall be controlled in accordance with MIL-STD-1523.

3.5.1.3 Metals. All metals shall be compatible with the intended temperature, functional, service and storage conditions to which the starter will be exposed. Metals which are not protected from corrosion by hydraulic fluid shall be of the corrosion-resisting type, or shall be suitably treated to resist corrosion in accordance with 3.5.1.4. Cadmium and zinc plating shall not be used for internal parts or on internal surfaces in contact with hydraulic fluid or exposed to its vapors.

3.5.1.3.1 Steels (except corrosion-resistant).

a. Aircraft quality, vacuum-melted steel shall be used for parts which are heat treated to an ultimate tensile strength 220,000 psi and above. Such applications shall be subject to specific approval by the qualifying activity.

b. The ultimate tensile strength range in production parts shall not exceed the following limits:

Minimum Ultimate Tensile Strength, P.S.I.	Permissible Variation, P.S.I.
220,000	-0 + 20,000
260,000	-0 + 20,000
270,000	-0 + 20,000
280,000	-0 + 20,000

c. Preference shall be given, in selection of carbon and low alloy steels, to compositions having the least hardenability which will insure through-hardening of the part concerned.

d. Compositions shall be selected such that heat treatment to the required strength and service temperatures shall preclude temper embrittlement.

e. Steels shall be selected having ductile-brittle fracture transition temperatures as determined by impact test below the minimum operating temperature.

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f. Steels whose mechanical properties are developed by cold deformation shall be so selected that the recovery temperature will be at least 50°F (10°C) above the maximum operating temperature range.

g. Critical parts shall be designed and processed so as to result in no decarburization of highly stressed areas. Designs shall preclude use of as-forged surfaces unless specific approval is obtained from the qualifying activity.

h. The mechanical drilling of holes in martensitic steels after hardening to strength levels of 180,000 psi and above shall be avoided whenever practicable. When such drilling is unavoidable, detailed information concerning the processes to be used shall be submitted to the qualifying activity for approval.

i. Any necessary straightening of parts after heat treatment to strength levels of 180,000 psi and above shall be accomplished at the tempering temperature, plus 0°F (-18°C) minus 50°F (10°C), or the parts shall receive a stress-relieving treatment at this temperature immediately after straightening. Parts shall be inspected for cracks after straightening.

j. Free-machining grades of steel shall be subject to approval by the qualifying activity.

3.5.1.3.2 Corrosion-resistant steels. The following limitations shall be observed in the selection and application of corrosion resistant steels.

a. Unstabilized austenitic steels shall not be fusion welded.

b. Precipitation hardening semi-austenitic grades shall not be used in applications which require extended exposure to temperatures in the 750° -900°F (399° -482°C) range.

c. The use of heat treated 431 or 19-9DL steel shall be subject to approval by the qualifying activity.

d. The use of 17-4PH stainless steel aged below H1025 shall be subject to approval by the qualifying activity.

3.5.1.3.3 Magnesium alloy. Magnesium or its alloy shall not be used.

3.5.1.3.4 Castings. Castings shall be in accordance with MIL-C-6021 and shall be classified Class 1B, grade C or higher. Aluminum alloy castings shall be in accordance with MIL-A-21180. Defects not

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materially affecting the suitability of the castings may be repaired at the foundry or during machining by peening, impregnation, welding, or other methods acceptable to the procuring activity. Inspection of repaired castings shall be governed by quality control techniques subject to the approval of the qualifying activity.

3.5.1.3.5 Stress corrosion embrittlement factors. To prevent premature failures caused by stress corrosion or hydrogen embrittlement, the design and method of manufacture of parts of titanium and steel heat-treated to tensile strengths above 220,000 psi or of bare high strength aluminum alloys and the techniques by which they are assembled and installed shall be such that sustained or residual surface tensile stresses and stress concentrations are minimized. Practices, such as: the use of press or shrink fits; taper pins, clevis joints in which tightening of the bolt imposes a bending load on the female lugs; and straightening or assembly operations; which result in sustained or residual surface tensile stresses shall be avoided. In cases where such practices cannot be avoided, corrective practices such as stress relief heat treatment, optimum grain flow orientation, peening (shot or rotary flat) or similar surface working shall be used to minimize the hazard of stress corrosion or hydrogen embrittlement damage. In no case shall sustained or residual surface tensile stresses in these materials exceed 50% of the material specification minimum yield strength in the longitudinal grain flow direction, 35% of the minimum yield strength in the long transverse direction and 15% of the minimum yield strength in the short transverse direction, except that for materials of demonstrated high resistance to stress corrosion cracking, such as 7075-T73 aluminum alloy, such stresses may be as high at 75% of the minimum yield strength in any grain flow direction. These limits for stresses represent the algebraic sum of all the continuous tension and compression surface stresses resulting from any source such as quenching, forming and assembly. The use of cadmium in contact with or plated on titanium shall not be permitted.

3.5.1.3.6 Fatigue factors. To prevent premature failures caused by repeated loads, the design and method of manufacture, including consideration of the damaging effect of decarburization and certain metallic coatings, of all critical parts and the techniques by which they are assembled and installed shall be such that sustained or residual tensile stresses and stress concentrations are minimized. Practices such as cold straightening, cold forming, and the assembly of mismatched surfaces, which result in sustained or residual surface tensile stresses shall be avoided. In cases where such practices cannot be avoided, corrective practices such as stress relief heat treatment, optimum gain flow orientation, peening (shot or rotary flat) or similar surface working shall be used to minimize premature fatigue failure. In no case shall parts of nonuniform section be cold formed unless the contractor can demonstrate with test results that required fatigue resistance is obtained.

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3.5.1.3.6.1 Surface finish. Surface roughness (as defined in ANSI B46.1-1962) and the incidence, location and size of allowable flaws in metallic parts subject to repeated loads, wherein stress concentration factors are governing, shall be demonstrably consistent with the design fatigue life of the parts in the planned operating environment. Surface roughness limits and limiting criteria for allowable flaws shall be established for approval by the qualifying activity. Alternately the surface roughness shall not exceed 63 RHS and there shall be no flaws such as tool marks deeper than 0.0002 inch. Such limits for parts critical in low cycle fatigue shall be separately identified.

3.5.1.3.7 Heat treatment. Heat treatment of aluminum alloys shall be in accordance with the material specification and MIL-H-6088. Steels shall be heat treated in accordance with the material specification and MIL-H-6875. Precautions shall be taken to minimize warpage during heat treatment. Steel parts which require straightening after hardening to 200,000 psi or below may be cold straightened provided a stress relieving heat treatment is subsequently applied. Straightening of parts hardened to tensile strengths above 200,000 psi shall be accomplished at a temperature within the range from the tempering temperature to 50°F (10°C) below tempering temperature.

3.5.1.3.8 Welding. Welding joints shall be in accordance with applicable welding specifications. Welded assemblies involving steels and section sizes which transform on air cooling to microstructures other than martensite shall be normalized or otherwise processed to equivalent hardness in the weld areas. When welding air hardening steels, appropriate precautionary measures, including preheating interpass temperature control and post heating shall be applied.

3.5.1.3.8.1 Stress corrosion. To avoid the possibility of stress corrosion and fatigue damage in welded primary structural components, the structures shall be so designed that the weld bead material is accessible for machining during manufacture. In addition, welded primary structural components shall be stress relieved after all welding is completed.

3.5.1.3.9 Corrosion protection. Metals which do not possess adequate corrosion-resisting characteristics shall be protected to resist corrosion from dissimilar metal combinations, moisture, salt spray and high temperature deterioration as applicable. The protection of the equipment from corrosion shall be in accordance with MIL-F-7179. Dissimilar metals in immediate contact shall be avoided. Where use of dissimilar metals in immediate contact cannot be avoided they shall be insulated. This has particular reference to dissimilar metal contacts defined in MIL-STD-889. All metal joints and other contact areas shall be designed or protected to prevent fretting corrosion damage of affected areas. The following factors will be used in the evaluation of the resistance of the equipment to corrosion. Any corrosion that causes the following shall be deemed sufficient cause for rejection.

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a. Malfunctioning of the equipment

b. Shortening of life

c. Impairment of use

d. Impairment of ease of replacement of parts

3.5.1.4 Finish.

3.5.1.4.1 Aluminum alloy parts. Unless otherwise specified by the qualifying activity, aluminum alloy parts shall be anodically treated in accordance with MIL-A-8625. In the absence of abrasive conditions and with the approval of the qualifying activity, they may be coated with chemical film in accordance with MIL-C-5541. The film deposited by this treatment shall be removed from contact areas required to act as a path for electrical current.

3.6 Design.

3.6.1 General. Hydraulic starters shall be designed such as to fulfill their intended purpose as specified in 1.2 and be of such construction as to satisfactorily withstand the expected normal usage encountered in service under the environmental conditions in 3.6.2 and as substantiated by tests specified in 4 of this specification.

3.6.2 Environmental conditions. The hydraulic starter shall be capable of being operated, without impairment of function or change in adjustment, throughout the ambient temperature range specified on the applicable specification sheet and shall not be damaged when exposed to temperatures specified thereon. Consideration shall be given in the design of the hydraulic starter to satisfactory operate during and after exposure to any of the following conditions in worldwide operation:

a. Humidity

b. Salt fog

c. Sand and dust

d. Fungus

e. Mildew

3.6.2.1 Acceleration forces. While mounted rigidly during non-operating periods, starters shall be capable of withstanding non-vibratory linear acceleration forces of 10 g applied at the starter center of gravity in any direction, without structural damage. Starters

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shall be capable of normal operation after exposure to this force. During starter operation, starters shall not generate acceleration forces of sufficient magnitude to exceed any of the design requirements of the respective engine accessory drive pad and mounting pad.

3.6.2.2 Vibration. Hydraulic starters shall be inherently balanced and designed to meet the vibration test requirements specified in 4.5.19.

3.6.3 Operating fluid. The hydraulic starter shall operate using hydraulic fluid conforming to MIL-H-5606 or MIL-H-83282. Temperature of the fluid shall be as encountered as the result of ambient temperature specified under 3.6.2 of this specification, but not higher than 275°F maximum. The acceptable contamination level shall be that provided by the continuous filtration of the starter inlet fluid through a 15-micron absolute filter conforming to MIL-F-8815.

3.6.4 Component parts. The starter shall consist of a hydraulic motor suitable for converting fluid energy into mechanical energy in the form of rotary motion. Variable displacement units shall be equipped with suitable devices to provide a controlled change of displacement during the operating cycle. In addition, the starter shall incorporate a suitable clutch for coupling the starter drive with the engine drive, a shear section or slip clutch in the output drive, and an automatically-operated cutout device. Devices for limiting input power and speed shall be supplied with Type I starters and when specified (see 6.2), for Type II starters.

3.6.5 Leakage.

3.6.5.1 Drive shaft seal. Where applicable, drive shaft seals shall be designed so that positive leakage, past the seal, shall not exceed the maximum leakage rates permitted in the applicable specification sheet.

3.6.5.2 Static seal. Leakage in the form of drops from the starter housing or from static seals shall not be permitted.

3.6.6 Part numbering of interchangeable parts. All parts having the same design activity FSCM (Federal Supply Code for Manufacturers) and part number shall be interchangeable as defined in MIL-STD-280. The item identification and part number requirements of DOD-STD-100 shall govern the design activity part numbers and changes thereto.

3.6.7 Threaded parts.

3.6.7.1 Straight screw threads. All conventional straight screw threads shall conform to the requirements of MIL-S-8879 except for electrical connections, thermocouple harness, fluid fittings and ground and cut threads incorporated in proprietary parts which do not have equivalent standards. All exceptions shall conform to MIL-S-7742.

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3.6.7.2 Tapered pipe threads. Tapered pipe threads shall be in accordance with MIL-P-7105 and may be used only for permanently plugged, drilled or cored openings.

3.6.7.3 Threaded inserts. All threaded connections in nonferrous materials shall have steel inserts which are suitably protected from electrolytic corrosion. Fill and drain boss inserts shall be designed to permit the use of standard gaskets or seals and standard straight thread plugs.

3.6.7.4 Safetying. Provisions shall be made to prevent all screws and screw parts from accidental loosening. Such parts shall be safety wired or fitted with safety stop nuts, where applicable. All other connections liable to be loosened by vibration shall be safetied in a similar manner. If safetying is by wiring, it shall be in accordance with MS33540. The use of lock nuts which incorporate nonmetallic inserts as a locking feature shall be avoided.

3.7 Filter. If filters are required, they shall be self-cleaning and shall be sized to protect the starter from the contamination that is present in the starter system. Any filter used shall be readily removable for replacement without disassembly of the starter.

3.8 Snapring retainer. The use of snapring retainers is prohibited unless failure of the retainer will not affect operation adversely or allow retainer fragments to enter the air passages.

3.9 Reliability and maintainability.

3.9.1 Reliability. The starter shall be designed and developed to achieve the highest operational reliability commensurate with the design requirements. Factors to be considered during the design and manufacture of the starter to assure in-service reliability are:

a. Simplicity of design.

b. Selection and application of reliable components.

c. Consideration of operational and environmental parameters.

d. Mechanical structures - adequate design stress margin on all parts.

e. Use of proven manufacturing techniques.

f. Rigid quality control procedures imposed throughout procurement, manufacturing, assembly, and testing to assure that maximum design reliability is built into the equipment.

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3.9.2 Maintainability. The starter shall have a minimum number of parts. Its design shall permit easy assembly, disassembly, location of trouble sources, routine service checking, adjustment and maintenance by personnel having a minimum of training. The modular concept of assembly shall be utilized wherever possible in order to increase maintainability and reduce overhaul and maintenance costs. The design goal shall be such that any subassembly of the starter can be replaced without disassembly of the complete starter or removal of other components. Maintenance between overhaul periods shall be held to an absolute minimum.

3.10 Functional requirements. The starters shall be designed and constructed to satisfy the functional requirements specified herein when operating under rated conditions and under conditions specified on the applicable specification sheet.

3.10.1 Rated conditions and cutout speed. Rated conditions and cutout speed for the starters are as defined herewith and are specified on the applicable specification sheet.

3.10.1.1 Rated inlet pressure. The rated inlet pressure is the maximum pressure required at the starter inlet port when operating at rated torque, rated outlet pressure, and rated temperature. The inlet pressure at zero fluid flow shall be as specified on the applicable specification sheet.

3.10.1.2 Rated outlet pressure. The rated outlet pressure is the pressure at the starter outlet port against which the starter is designed to produce rated torque at rated inlet pressure and rated temperature.

3.10.1.3 Rated inlet fluid temperature. The rated inlet fluid temperature is the maximum fluid temperature at the starter inlet port when operating at rated torque and rated pressures.

3.10.1.4 Rated fluid flow. The rated fluid flow is the maximum fluid flow needed to meet the torque requirements at any starter drive speed from zero rpm to cutout speed.

3.10.1.5 Rated torque. The rated torque is the minimum torque at the starter drive when operating at rated pressures, rated temperature, and rated speed.

3.10.1.6 Rated speed. The rated speed is the maximum starter drive speed at which rated torque is produced and is usually related to engine light-off speed.

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3.10.1.7 Cutout speed. The cutout speed is the starter drive speed at which the cutout device will signal the termination of starter operation.

3.10.2 Engagement. Engagement of the starter drive with the engine accessory drive shall be accomplished with a suitable clutch engagement, with minimum shock, and independently of any external signal, provided sufficient hydraulic energy is available to cause rotation of the hydraulic starter.

3.10.3 Disengagement and re-engagement. Disengagement shall be rapid and complete at the end of a starting cycle. Re-engagement of the starter with the engine shall be accomplished rapidly and without damage to starter or engine even though the starter or engine is rotating. Disengagement and re-engagement of the output shaft shall be independent of any external signal.

3.10.4 Rotation. The rotation of the starter drive shall be as specified on the applicable specification sheet.

3.10.5 Torque limiting device. Starters shall incorporate a slip clutch or shear section to prevent damage to the engine in the event of excessive starter torque. The output drive shall be designed so that if failure of the output drive occurs, the resulting parts will be retained within the starter and engine accessory drive.

3.10.5.1 Shear sections. The shear section shall be designed to permit replacement without disassembly of the starter and to limit axial movement of the output shaft. The maximum permissible static shear torque to be transmitted to the engine by the starter drive under all operating conditions, including engagement and re-engagements, shall not exceed the value specified on the applicable specification sheet.

3.10.5.2 Slip clutch. Slip clutch, when provided, shall have a maximum breakaway torque that will prevent the starter from transmitting torques to the engine starter drive in excess of the values specified in 3.10.5.1 under all conditons specified herein.

3.10.6 Lubrication. The starter shall be self-lubricating with hydraulic fluid. Overrunning clutches shall be designed to incorporate adequate lubrication to permit continuous operation at the conditions specified in 4.5.13 at the attitude and environmental conditions herein specified in 3.13.2.3 and 3.6.2, respectively. Only the specified hydraulic fluid shall be used to lubricate seals during assembly of the starters.

3.10.7 Automatic shutoff control. The starter shall be equipped with a cutout switch that will function in the manner and at a starter output shaft drive speed specified on the applicable

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specification sheet. The switch shall be designed to withstand the electrical load specified on the applicable specification sheet. The switch cutout speed setting shall be externally adjustable on the assembled starter and when the starter is installed on the mating accessory mounting pad. The switch shall not be polarity sensitive.

3.10.8 Overrunning. The starter shall not be damaged as a result of overrunning under the conditions specified on the applicable specification sheet with the maximum misalignment allowed for the engine accessory drive.

3.11 Performance.

3.11.1 Output. The starter shall meet the output torque characteristics at rated inlet hydraulic fluid temperature as specified on the applicable specification sheet.

3.11.2 Overspeed limit. Type I starters shall be designed to prevent overspeed when subjected to unlimited flow with the cutout speed switch inoperative. Type II starters shall be designed to prevent overspeed when subjected to the flow rate as specified on the applicable specification sheet with the cutout speed switch inoperative. The no-load speed of the output shaft of the starters at the above conditions shall be as specified on the applicable specification sheet.

3.11.3 Hydraulic fluid flow. When operating at rated inlet and outlet conditions, the fluid flow shall be as specified on the applicable specification sheet at the speeds indicated thereon.

3.11.4 Pressure application. The rate of pressure application to the starter shall be as specified in Figure 1 herein.

3.11.4.1 Surge pressures. During operation, starters shall be capable of withstanding inlet surge pressures not to exceed 4500 psig and outlet surge pressures not to exceed 400 psig. Inlet and outlet surge pressures may occur simultaneously or separately and shall not exceed one second in duration.

3.11.4.2 Proof pressure. At rated speed, torque and inlet temperature, the starter shall be designed to operate at the inlet pressure and outlet pressure specified on the applicable specification sheet.

3.11.5 Sustained motoring. At rated inlet conditions, the starter shall produce and maintain rated torque and speed for the time specified on the applicable specification sheet.

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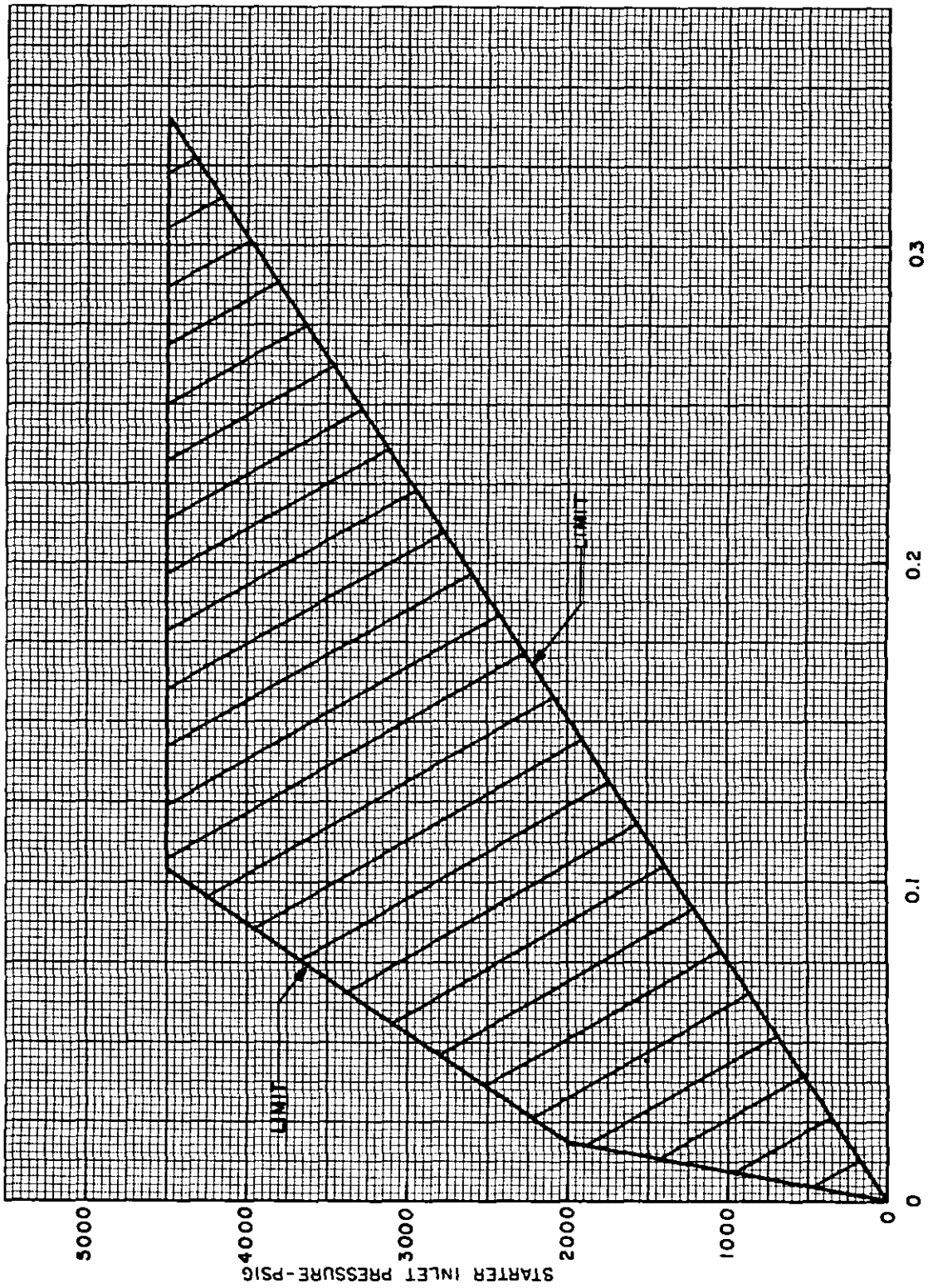


FIGURE 1. Starter inlet pressure rise time.

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3.11.6 Succession of starts. There shall be no limitation of sequence or number of starts performed in succession at starter rated conditions. The starter shall meet the endurance test specified in 4.5.7.

3.11.7 No-load speed. The starter shall be capable of running without damage at rated inlet conditions for the period specified on the applicable specification sheet with the starter output shaft unloaded and the cutout switch inoperative.

3.12 Weight. The weight of the hydraulic starter designed to this specification shall be as specified on the applicable specification sheet.

3.13 Installation.

3.13.1 Size and dimensions. The size of the starter shall be within the envelope configuration specified on the applicable specification sheet.

3.13.2 Mounting. The starter shall be designed for mounting on the engine accessory drive specified on the applicable specification sheet. Mounting or removal of the starter shall be accomplished as a single unit without separable parts and shall not require special tools.

3.13.2.1 Quick-Attach-Detach (QAD) mounting. Except for MIL-S-22999/1 starters, all starters shall be provided with quick-attach-detach mounting provisions which meet the requirements specified on the applicable specification sheets.

3.13.2.2 Engine accessory drive. The starter shall be suitable for mounting on the engine accessory drive specified on the applicable specification sheets.

3.13.2.3 Attitudes. Starter operation shall be satisfactory in any attitude as long as provision is made for the appropriate drainage of the shaft seal chamber.

3.13.2.4 Mounting pad drain. The hydraulic starter shall incorporate a mounting pad port to drain fluid as a result of leakage from the output drive shaft seal and engine accessory drive shaft seal. The mounting pad drain port shall conform to MS33649. The port size and location shall be shown on the manufacturer's approved installation drawing.

3.13.3 Ports. The inlet and outlet ports shall conform to MS33649. No case return port is permitted. The starter must incorporate internal provisions to allow fluid entering the case to be returned to the starter discharge port. The size of the ports shall be as specified on the applicable specification sheet. The port size and location shall

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be shown on the manufacturer's approved installation drawing. The ports, except for the MIL-S-22999/1 starter, shall be of different sizes to eliminate interchanging connections and shall be clearly and permanently identified. The inlet port shall be marked HP; the outlet port, LP.

3.14 Marking.

3.14.1 Nameplate. Each hydraulic starter shall be clearly and permanently marked with an irreversible nameplate in accordance with MIL-STD-130, securely attached to a part of the starter which will not ordinarily be renewed during normal service life, specifying the following information:

STARTER, AIRCRAFT ENGINE, HYDRAULIC, FOR _____ ENGINE
Specification MIL-S-22999(AS)
Part number M22999/ (Insert applicable slash number)
*Rotation (Indicate by an arrow)
Manufacturer's part number _____**Less issue _____
Manufacturer's serial number
Government order or contract number
National Stock Number

NOTES: *The arrow showing the direction of rotation may be included on a single nameplate with other specified data or, at the option of the contractor, it may be stamped on a separate plate attached immediately adjacent to the main nameplate.

**For use by Naval Air Rework Facilities.

3.14.2 Marking of product. The Navy model designation shall not be used on starter until notification has been received in writing that the starter has been approved for use in Naval aircraft.

3.14.3 Marking of parts. Each part and assembly shall be marked with its part number, and drawing revision symbol where applicable, which shall be the same as the drawing number, except for the following:

3.14.3.1 Those parts which do not have a suitable or sufficient surface for a part number.

3.14.3.2 Those parts which are permanently assembled by welding, brazing, soldering, or riveting shall carry the assembly part number.

3.14.3.3 Part number. The part number shall, when practicable, be located to permit its being read after assembly of the complete unit. The marking shall be such that it will not be effaced or obliterated as a result of service usage during the life of the part.

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3.15 Part number and parts list.

3.15.1 Part numbers. Starter part numbers are not to be coded to describe part characteristics and shall be limited to a maximum of 10 digits or characters. All parts shall be marked with the latest drawing revision letter or number to which the part was made.

3.15.2 Parts list. A parts list is a tabulation of the items necessary to fabricate or assemble the starter to which the list applies. The parts list for the starter which is granted qualification approval based on (1) the satisfactory completion of the qualification tests and (2) the approval of the qualification test detail inspection and test report by the qualifying activity shall consititute the approved parts list for any subsequent starter of the same model to be delivered to the Government. All parts shall be marked directly with an identification number and the vendor or fabrication source of that particular part, unless otherwise specified in the parts list. The contractor shall assign an item configuration control number to the starter configured by the approved parts list, and this number together with the approved parts list shall be continuously updated to reflect the incorporation of Class I or Class II engineering changes approved by the qualifying activity. This item configuration control number will be used to identify the starters to be delivered and will also be used for final acceptance. The approved parts list shall be prepared in accordance with Chapter 3 of DOD-STD-100 and this list shall also show the latest drawing revision letter or number used to manufacture the part incorporated in the starter at the time of acceptance. The use of the so-called "clip system" of attaching engineering change forms to the applicable drawing in lieu of showing the change(s) in the body of the drawing will not be permitted.

3.15.3 Design and construction changes.

3.15.3.1 Changes in design. No changes shall be made in the design or materials of a starter that has been qualified in accordance with this specification, except when the changes are approved by the qualifying activity.

3.15.3.2 Changes in part numbers. Part number changes shall be controlled in accordance with 1-302.14 of DOD-STD-100.

3.15.3.3 Changes in the parts list. Changes to the approved parts list shall be governed by the requirements specified in 3.15.3.2.

3.15.3.4 Changes in vendors. Changes in a vendor or fabrication source of those parts used in the qualified starter shall be in accordance with the following procedure: The manufacturer shall prepare and submit to the qualifying activity a list of tests to demonstrate that the starter meets all specification requirements for an alternative

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vendor source. The specific tests required to substantiate an alternate vendor as a supplier of a starter part shall be defined. The fabrication source of selected vendor components will be included in this list. The list shall be subject to approval by the qualifying activity. The contractor thereafter shall be responsible for insuring that all parts, components and assemblies on the substantiation list comply with the qualified fabrication source, and that any changes to those sources are effectively controlled. The contractor shall be responsible for performance of substantiation tests to establish satisfactory alternate vendors or fabrication sources. A fabrication source is defined as the prime physical source producing the part, component, or assembly. Changes of fabrication location, such as to another plant of an individual vendor, shall be construed as a change of fabrication source.

3.15.3.5 Responsibility for changes. Approval by the qualifying activity of changes to the qualification test starter does not relieve the manufacturer of full responsibility for the results of such changes on any starter characteristic.

3.16 Workmanship. The workmanship and finish shall be of sufficiently high grade to insure satisfactory operation, reliability, and durability consistent with the application and storage life requirements of the starter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the hydraulic starter shall be classified as follows:

a. Qualification tests (4.3)

1. Retention of qualification - Retention of qualification consists of a periodic verification to determine compliance of the qualified product with the requirements of this specification (see 3.2 and 4.3).

b. Quality conformance inspection (4.4)

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4.3 Qualification tests.

4.3.1 Qualification test samples. The qualification test sample shall consist of two hydraulic starters and one additional output shaft coupling per starter of each manufacturer's part number upon which qualification is desired. These hydraulic starters and one additional output shaft coupling per starter shall be accompanied by one complete parts list, one complete set of detail and assembly drawings and a qualification test report. The drawings shall be in accordance with DOD-STD-100. Each qualification test sample shall be plainly identified by securely attached tags clearly and durably marked with the following information, and shall be forwarded to the Commanding Officer, Naval Air Propulsion Center, Attn: Code PE-61, Trenton, New Jersey 08628, after obtaining authorization from the qualifying activity for submittal of the qualification test samples:

Sample for Qualification Test
Submittal by (name) (data) for qualification test
in accordance with the requirements of
Specification MIL-S-22999 (date) under
authorization (reference letter authorizing
tests).

Starter, Aircraft Engine, Hydraulic, General
Specification for
Manufacturer's Part Number
Name of Manufacturer
Specification Sheet Model or other identification
Navy Work Unit Assignment Number

4.3.1.1 Qualification test report. The supplier shall furnish performance data on each model starter they propose to have qualified to the qualifying activity in accordance with the requirements of 4.3.1 herein. The data shall be in the form shown on Figure 2 and may be estimated or calculated until the starter has been tested. After that time, performance data shall be based on test results. The report shall demonstrate that the size of the starter is within the envelope configuration specified on the applicable specification sheet. The starter installation configuration shall be as shown thereon and shall be subject to approval by the qualifying activity. Manufacturers interested in qualifying starters to this specification are required to coordinate installation with the qualifying activity prior to fabrication of qualification units. The manufacturer's approved installation drawing shall form a part of this specification.

4.3.1.1.1 Manufacturer's drawings. The manufacturer shall furnish the following drawings and diagrams to the qualifying activity with the submission of the qualification test samples.

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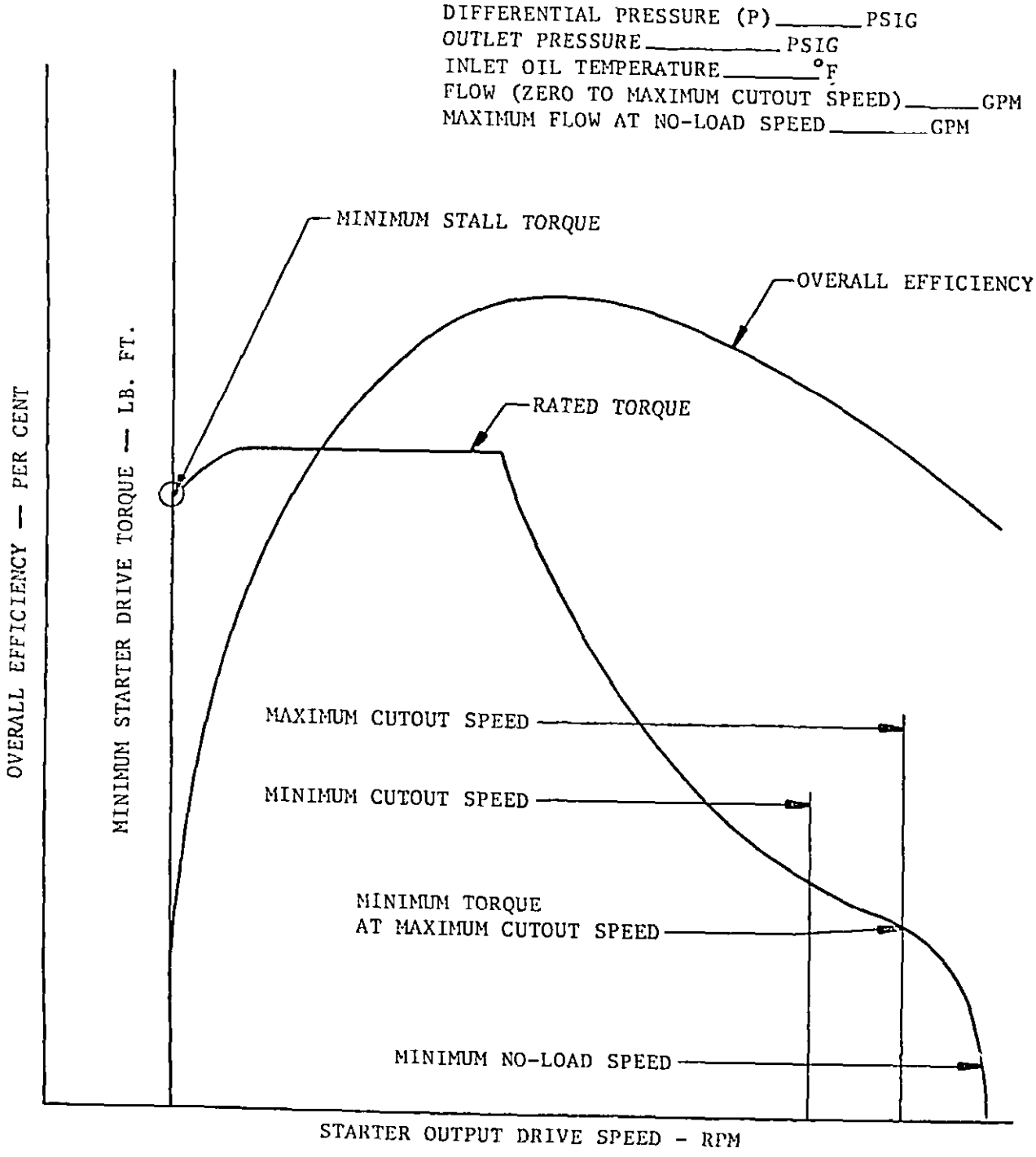


FIGURE 2. Minimum Starter Drive Torque and Overall Efficiency Versus Starter Output Drive Speed

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a. Hydraulic starter installation drawing - The drawing shall show hydraulic starter center of gravity, all connecting fittings, clearance required for installation and removal, filter locations and cleaning instructions if applicable.

b. Electrical diagram - The electrical diagram shall show all electrical circuits up to the connecting points of the aircraft electrical system.

c. A complete cutaway or cross sectional drawing of the starter and all components showing all parts in their normal assembled position and shall specify part numbers of all parts or components subassemblies. The following data shall be furnished on or together with the assembly drawing:

1. All dimensions, clearances, and fits

2. Materials, treatment and finishes (identify each by specification number)

3. Construction

4. Characteristics and ratings

d. A complete set of manufacturing drawings and an applicable parts list.

e. Packing requirements and packing drawings for the starter.

f. All contractor specifications listed in the manufacturing drawings.

4.3.2 Tests. The qualification tests shall consist of the tests specified under 4.5 and listed in Table I, conducted on the applicable specimens. The tests to be conducted on each starter shall be determined by the qualifying activity. If considered desirable by the qualifying activity, the qualification test may be supplemented with tests under actual or simulated service conditions, provided they are not in contradiction to the requirements specified herein. The qualifying activity reserves the right to disassemble the starters partially or completely before, during, and after the qualification test.

4.3.3 Retention. The retention of qualification shall consist of certification by the manufacturer to demonstrate compliance of the qualified starter with the requirements of this specification. Certification shall be signed by a responsible official of management, attesting that the listed product(s) is still available from the listed plant, can be produced under the same conditions as originally qualified; i.e., same process, materials, construction, design, manufacturer's part number, or designation; and meets the requirements of the current issue of the specification. Certification shall be conducted at intervals not exceeding 2 years.

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TABLE I. Qualification Tests

Test	Paragraph
Examination of product	4.5.3
Initial calibration	4.5.4
No-load operation	4.5.5
Proof pressure test	4.5.6
Endurance test	4.5.7
Post-endurance calibration	4.5.8
Leakage (operational)	4.5.9
Leakage (static)	4.5.10
Sustained motoring	4.5.11
Stall torque	4.5.12
Overrunning	4.5.13
Extreme temperature operation	4.5.14
Consecutive operation	4.5.15
Dynamic torque check	4.5.16
Disengagement and re-engagement	4.5.17
Shock test	4.5.18
Vibration	4.5.19
Shear section strength	4.5.20
Engine cranking	4.5.21
Teardown inspection	4.5.22
Over-pressurization	4.5.25

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4.4 Quality conformance inspection. Quality conformance inspection shall consist of the following:

Individual inspection (4.4.1)

4.4.1 Individual inspection. The individual inspection of the hydraulic starter shall consist of the following tests conducted in the order specified:

a. Torque calibration and flow (4.5.23).

b. Overrunning (4.5.13 except the duration of the test shall be for a 30-minute period at the maximum speed specified).

c. Automatic shutoff control (4.5.7 except the duration shall be for 3 complete cycles only).

d. Leakage static and shaft seal leakage (4.5.10 and 4.5.24).

e. Proof pressure test (4.5.6).

f. Examination of product (4.5.3).

g. Shaft seal leakage (4.5.24).

In addition to the previously mentioned tests, the starters may be subjected to any other tests specified herein, recommended by the Government Inspector and authorized by the Navy to determine conformance with the requirements of this specification.

4.5 Tests.

4.5.1 Test conditions. Unless otherwise specified in this specification or the applicable specification sheet, all tests shall be conducted with the starter at an ambient temperature range of 50° to 120°F, with rated conditions as specified on the applicable specification sheet. Tests may be conducted in the order deemed most desirable by the qualifying activity.

4.5.1.1 Pressure rate increase. Unless otherwise specified herein, the pressure rate increase during all tests shall be within the limits of Figure 1 herein.

4.5.1.2 Filtration. For all tests except the low temperature test, an absolute 15-micron filter shall be installed in the starter inlet and discharge hydraulic fluid lines. The filter shall be in accordance with MIL-F-8815. The filter shall be inspected after every 100 starts and after every 100 hours of overrunning. Excessive accumulation of chips or other foreign material originating from the starter shall be cause for rejection.

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4.5.1.3 Hydraulic fluid. Fluid conforming to MIL-H-6083 may be used during acceptance test at the discretion of the contractor.

4.5.2 Instrumentation. Instrumentation shall be provided during qualification and inspection tests to observe or determine the following:

4.5.2.1 Outlet pressure. The outlet pressure shall be measured with a calibrated pressure gage located within two inches of the outlet port. Gage accuracy shall be within ±1 percent psig.

4.5.2.2 Inlet pressure. The inlet pressure shall be measured with a calibrated pressure gage located within two inches of the inlet port. Gage accuracy shall be within ±1 percent psig.

4.5.2.3 Hydraulic flow. The flow from the outlet shall be measured by any method which will give results within an accuracy of +2 percent.

4.5.2.4 Speed. The speed shall be measured within an accuracy of ±1 percent of output shaft speed.

4.5.2.5 Torque. The torque developed by starter shall be measured by a suitable method approved by the Navy within an accuracy of ±1 percent.

4.5.2.6 Fluid temperature. Hydraulic fluid temperature shall be measured within an accuracy of ±5°F.

4.5.2.7 Ambient temperature. Ambient temperature shall be measured in the vicinity of the starter with an accuracy of 1 percent of maximum reading.

4.5.3 Examination of product. Each hydraulic starter shall be subjected to a careful inspection to insure all external dimensional tolerances are within specification and drawing limits and that all parts are correctly assembled.

4.5.4 Initial calibration. Initial calibration shall consist of starter operation at starter drive speeds ranging from zero to free-running speed and observing the parameters listed in 4.5.2. A minimum of six stabilized speeds shall be obtained. The starter oil flow versus speed curve shall be within the limits specified on the applicable specification sheet. The starter performance shall be within the limits specified on the applicable specification sheet. External cooling may be used during this calibration if excessive oil temperatures will result from prolonged operation. For Type I starters, this test shall be conducted at the two pressure levels noted on the applicable specification sheet.

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4.5.5 No-load operation. The starter shall be operated at no-load, with the cutout switch inoperative, without damage or failure for the time specified on the applicable specification sheet. The starter free-running speed obtained during this test, or the maximum speed attained before automatic cutback, shall not be less than that specified on the applicable specification sheet.

4.5.6 Proof pressure test. With the loading device adjusted to limit starter output drive speed to rated speed rpm and at rated inlet temperature, the starter shall be operated at the conditions noted on the applicable specification sheet. Shaft seal leakage shall not exceed the rate specified on the applicable specification sheet.

4.5.7 Endurance test. The starter shall be subjected to an endurance test as specified on the applicable specification sheet without damage to, adjustment of, or replacement of any of the component parts. Each cycle shall consist of the phases specified on the applicable specification sheet. The cutout switch shall be electrically loaded to the value noted on the applicable specification sheet for each cycle and shall function within the limits specified thereon.

4.5.8 Post-endurance calibration. After completion of the endurance test specified in 4.5.7, the rated torque speed test of 4.5.4 shall be repeated. The starter overall efficiency shall not have decreased more than 5 percent from the original values at rated load.

4.5.9 Leakage (operational). The shaft seal leakage shall be determined during the endurance test of 4.5.7 herein and shall not exceed a rate of 5 cc per 120 endurance cycles. Other external leakage shall be of insufficient magnitude to form a drop.

4.5.10 Leakage (static). With hydraulic fluid at rated temperature and at rated outlet pressure and while the inlet port is blocked, external leakage and shaft seal leakage shall be of insufficient magnitude to form a drop when measured over a 15-minute period.

4.5.11 Sustained motoring. While coupled to a loading device, the starter shall be operated at rated inlet conditions, rated speed, and rated torque load for the time period specified on the applicable specification sheet. No damage shall result from this operation.

4.5.12 Stall torque. The stall torque of the starter shall be measured at rated inlet and outlet conditions at four equally-spaced positions within one revolution. The minimum torque measured shall not be less than the torque specified for zero speed in 4.5.4.

4.5.13 Overrunning. The starter output shaft shall be driven for the time period specified and in accordance with the conditions noted on the applicable specification sheet. The starter motor case is

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to be completely filled with MIL-H-5606 hydraulic fluid and the test system containing at least two gallons of fluid with the case pressurized to rated outlet pressure. One pint of fluid may be drained from the starter after each 24 hours of overrunning; one pint of fluid may then be returned to the system. No damage to the starter shall result from this test.

4.5.14 Extreme temperature operation.

4.5.14.1 Low temperature operation. The starter shall be stabilized at -65°F. After fluid and starter have reached stabilized temperatures, the starter shall be operated at rated inlet and outlet pressures and the fluid at rated inlet and outlet pressures. It shall then be operated for three consecutive cycles without damage. Each cycle shall be equivalent to one cycle as described for the endurance test in 4.5.7 except that the starter output shall meet the requirements specified on the applicable specification sheet. Fluid temperature rise shall be recorded during the three consecutive cycles.

4.5.14.2 High temperature test. The starter shall be stabilized at an ambient temperature of 275 ±10°F. The starter shall be operated for three consecutive cycles at rated pressures and rated fluid temperature without damage. Each cycle shall be equivalent to one cycle as described for the endurance test in 4.5.7.

4.5.15 Consecutive operation. The starter shall be operated for 10 consecutive endurance cycles without damage. Each cycle shall be equivalent to one cycle as described for the endurance test in 4.5.7.

4.5.16 Dynamic torque check. Cycling as described in 4.5.7 shall be run, but using continuous torque recording instrumentation. Response rate of the instrumentation used shall have a flat sinusoidal response to 1000 cps. The peak transient torque shall not exceed the value listed on the applicable specification sheet.

4.5.17 Disengagement and re-engagement test. Disengagement and re-engagement tests shall be conducted on the starter to demonstrate compliance with 3.10.3 herein. The starter shall be disengaged and re-engaged at speeds in increments of 1000 rpm from zero rpm to cutout speed. No damage to the starter shall result from this test. The recording instrumentation of 4.5.16 shall be used to measure impact torques at re-engagement.

4.5.18 Shock test. The starter shall be subjected to the shock test in accordance with Method 516.2, Procedure I of MIL-STD-810 using Figure 516.2-1 at amplitude "a". As a result of the shock test the overall efficiency of the starter shall not have deteriorated more than one (1) percent.

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4.5.19 Vibration. A vibration test shall be conducted in accordance with MIL-STD-810, Method 514.2 and as follows:

a. Equipment category - b.1

b. Table - 514.2 - II

c. Procedure - 1

d. Figure - 514.2-2

e. Test curve - F

No damage to the starter shall result from the test.

4.5.20 Shear section strength. After completion of all tests, the output shaft couplings of the two qualification starters submitted in accordance with 4.3.1 herein shall be sheared and the shearing section torque recorded. The shear value shall not exceed the values specified on the applicable specification sheet.

4.5.21 Engine cranking tests. At the option of the qualifying activity, the starter must complete 75 actual starts of the engine for which it was designed. The starter may be one submitted for qualification or it may be an additional model furnished by the starter manufacturer for such tests. The tests may be performed in an engine test cell at the Naval Air Propulsion Test Center, Trenton, New Jersey 08628, at the engine manufacturer's plant, in a prototype aircraft, or in an aircraft specifically designed for such tests. Test conditions shall be at the prevailing ambient conditions where the tests are performed and the inlet and outlet pressure shall be as specified in 3.10.1.1 and 3.10.1.2.

4.5.22 Teardown inspection. A teardown inspection consisting of complete disassembly of the starter shall be made at the completion of the qualification test. Dimensional inspection shall be made of all wearing parts. Rotating parts shall be subjected to Magnaflux or Zyglo inspection, except that assembled anti-friction bearings shall not be subjected to Magnaflux inspection. Rotating parts subjected to centrifugal stresses shall be measured on dimensions likely to show growth. Where practicable, seal clearances shall be determined. Failure of or any abnormal wear or damage to any part, sufficient to impair continued use of the starter, shall be sufficient cause for rejection of the starter.

4.5.23 Torque calibration and flow (quality conformance only). A calibration of torque versus speed and fluid flow shall be conducted to determine that the starter meets the minimum starter drive torque and flow requirements specified on the applicable specification sheet.

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4.5.24 Shaft seal leakage (quality conformance only). With the inlet port blocked off and the outlet port pressurized at rated outlet pressure, the starter shall be overrun for a period of 30 minutes at the drive speed specified on the applciable specification sheet. This leakage check may be conducted during the overrunning test specified in 4.4.1(b).

4.5.25 Over-pressurization. The starter shall be operated at rated conditions except for the inlet pressure which shall be as specified on the applicable specification sheet. The starter shall complete 10 cycles in accordance with the endurance test specified in 4.5.7 without damage to or replacement of any of the component parts.

4.5.26 Rejection and retest. Any starter failing to meet the requirements of the qualification tests shall be rejected and returned at the contractor's expense. Starters which have been rejected may be replaced or repaired to correct the defects and resubmitted for all specified tests. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the original defects shall be furnished. Starters rejected after retest shall not be resubmitted without the specific approval of the qualifying activity.

5. PACKAGING

5.1 Application. The requirements specified herein apply only to direct purchases by or direct shipments to the Government.

5.2 Preservation-packaging. Unless otherwise specified by the contract or purchase order, all Class I starters shall be completely filled with undiluted hydraulic fluid conforming to MIL-H-6083 prior to sealing. All ports shall be sealed by closures conforming to MIL-C-5501. All exterior surfaces of the component shall be protected by corrosion-preventative compound conforming to MIL-C-11796 or MIL-C-16173. The component shall then be wrapped or bagged in grade A greaseproof paper conforming to MIL-B-121 and sealed with tape conforming to PPP-T-60.

5.2.1 Level A. Unless otherwise specified, each starter shall be packaged for shipment in accordance with MIL-P-116. Method IIb of this specification shall be employed for all starters designated as installation starters. Method IId of this specification shall be employed for all starters designated as spare starters.

5.2.2 Level C. Starters shall be packaged in accordance with standard commercial practice.

5.3 Packing. Shipping containers, insofar as practicable, shall contain an identical number of articles, shall be of a uniform size, and shall be designed to enclose the contents in a snug, tight-fitting manner. When packed for shipment, the gross weight of the exterior container shall not exceed 200 pounds. The above applies to 5.2.1 and 5.2.2.

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5.3.1 Level A. Starters packaged in accordance with 5.2.1 shall be packed in an exterior shipping container in accordance with PPP-B-601 (overseas type) or PPP-B-621 (overseas type).

5.3.2 Level B. Starters packaged as specified in 5.2.1 shall be packed in an exterior shipping container conforming to PPP-B-601 (domestic type) or PPP-B-585.

5.3.3 Level C. The interior packages shall be placed in substantial commercial exterior shipping containers constructed to insure acceptance by common or other carrier for safe transportation at the lowest rate, to the point of delivery. Except as specified herein, the container shall conform to the requirements of the Uniform Freight Classification Rules in effect at the time of shipment.

5.4 Marking. Each package and exterior shipping container shall be marked to comply with MIL-STD-129 and to include the following:

STARTER, AIRCRAFT ENGINE, HYDRAULIC, FOR _____ ENGINE
Specification MIL-S-22999
Specification Sheet MIL-S-22999/*
Mfg. Model Designation
Ser. No.
Date of Manufacture
Contract No.
NSN

NOTE: *Insert applicable slash number.

6. NOTES

6.1 Intended use. These starters are intended for use in starting aircraft gas turbine engines.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, date of this specification, and applicable specification sheet title and part number.

b. Selection of applicable levels of preservation, packaging and packing (5).

c. Power and speed limiting device, Type II starters (3.6.4).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for

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inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. A prorated share of the qualification testing conducted in a Government testing laboratory is to be borne by the offeror. The activity responsible for the Qualified Products List is the Naval Air Systems Command, Attn: AIR-53641, Department of the Navy, Washington, D.C. 20360; however, information pertaining to qualification of products may be obtained from the Commanding Officer, Naval Air Propulsion Center, Attn: (Code PE 61) Trenton, New Jersey 08628.

6.3.1 Mockup. The supplier shall conduct a mockup or trial installation to verify form, fit, and function of the product as required by this specification and to demonstrate compatibility with the applicable aircraft. Prior to submitting the starters to the Government for qualification testing, the suppliers should arrange, with the Naval Air Systems Command (AIR-536), a mockup or trial installation of the product that is offered to the Navy for qualification testing. Any costs incident to the actual mockup or trial installation on the aircraft is to be borne by the offeror.

Preparing activity:
Navy - AS
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